



9 layers, L1 → L9 8 enterprise controls 138 products assessed 1,449 sources 58 Strategic · 67 Tactical · 13 Experimental

CONTROL PLANE firm-owned policy and one evidence store, around every call

MODEL PLANE

L1 Foundation-model portfolio
two unrelated vendors + small + open-weight, all pinned

- A two-vendor mid tier qualified on one suite, a small tier and a self-hosted open-weight tier; every version pinned.

OpenAI GPT-6 Astra · Sol · Luna · 6.1 Sol	Mistral Medium 3.5 · Large 3 · EU-hosted	Anthropic Claude hyperscaler EU route · tier set by reader	Gemma 4 small open-weight tier · Apache 2.0
Google Cloud Google Gemini 3.x pin versions · short lifetimes	Grok 4.7 (SpaceX AI) via a hyperscaler only	DeepSeek V4 weights or in-tenant only	Z.ai GLM-5.3 MIT Flash weights · after sanctions review
Meta Muse / Llama block the contributor tier	Qwen 3.8 self-host by policy	Kimi K3 custom licence	

L2 Inference & model access
in-region access; vLLM as the private exit route

- The primary cloud's model service in an approved UK/EU region; one open-weight model on vLLM as the tested exit route.

PATTERN · NOT SCORED Primary cloud's model service Bedrock · Foundry · Gemini Enterprise Agent Platform, in region	vLLM default engine · the exit route	Fireworks AI once ISO certificates confirmed	SGLang backup engine once CVE is fixed
Hugging Face Hub governed open-weight supply	Together AI EU dedicated, ZDR on	OpenRouter behind the gateway · Stripe deal	Ollama developer tier
Cerebras low latency, non-confidential	LM Studio desktops only · no service use	Ilm-d CNCF sandbox · pilot	NVIDIA Dynamo beta · pilot

L3

Orchestration: workflows & agents

deterministic by default, durable, approval gates

Workflows by default with one bounded model step, durable execution, and approval interrupts only a named human can resume.

LangGraph

default · workflows + agents

Temporal

durable execution

Microsoft Agent Framework

GA · Microsoft and .NET estates

Pydantic AI

typed agent steps

Google Cloud

Google ADK

on Agent Engine

AWS

Strands + AgentCore Runtime

framework + managed runtime

OpenAI Agents SDK

sandboxed sub-step · pre-1.0

CrewAI

use Flows for regulated work

LlamaIndex

retrieval toolkit

Vercel AI SDK

TypeScript app tier

Claude Agent SDK

Alpha · sandboxed only · alt: LangGraph

Mistral Agents API

Workflows in beta

L4

Tools & connectivity

nothing reachable except through the governed gateway

Read-only tools for authoritative data behind a tool-governance sub-layer: private registry, pinned definitions, policy, audit.

A2A 1.0

cross-team delegation · signed cards

MCP

behind a governed gateway · alt: OpenAPI

AWS

AgentCore Gateway + Identity

managed tool governance

Browserbase

only where no API exists

E2B

microVM sandbox

Agent Skills

script-free, internal · alt: AGENTS.md

Exa

via egress proxy + DLP

Tavily

Nebius-owned

Composio

May 2026 token incident · self-host

KNOWLEDGE PLANE

L5

Memory service (part of L6)

policy-gated writes, erasure by person

A governed record class stored in L6: write gate, subject index, erasure and a snapshot per run. Built last.

AWS

AgentCore Memory

inside its own runtime

Cognee

in-estate only

Google Cloud

Memory Bank

Agent Engine

Supermemory

proprietary memory + knowledge API

Zep / Graphiti

Graphiti self-hosted

Letta

agent harness

Mem0

OSS behind a firm memory API

LangMem

no release since Oct 2025

L6

Retrieval & knowledge stores

derived, entitlement-filtered, rebuildable index

Hybrid search inside a platform you already run; a dedicated vector engine only when a load test proves the need.

Elasticsearch

hybrid + document-level security

Milvus / Zilliz

very large corpora

PostgreSQL + pgvector

where Postgres is standard

Qdrant

dedicated engine after a load test

MongoDB Vector Search

GA self-managed too

Pinecone

managed · BYOC

Weaviate

licence in transition

S3 Vectors

AWS cost tier · no BM25

turbopuffer

many tenants · BYOC only

Chroma

prototypes and harnesses

L7

Retrieval optimisation

pinned embed + rerank, hybrid fusion, eval gate

Choose models by in-domain evaluation; pin every version; keep raw text so a model switch is a re-embed, not a rebuild.

Sentence Transformers

self-hosting and fine-tuning toolkit

Jina AI

Elastic-owned · CC-BY-NC weights

Google Cloud

Gemini Embedding 2

EU endpoint excludes the UK

Qwen3 Embedding

self-host after review

Cohere Embed 5 + Rerank

private deployment

Voyage AI

MongoDB-owned

OpenAI text-embedding-3

text baseline

NVIDIA NeMo Retriever

NVIDIA estates

L8

Ingestion & data preparation

approved sources, ACLs and lineage on every chunk

A built control envelope (source register, classification, parse manifest, lineage, incremental indexing) around replaceable parsers.

Docling

default engine · LF AI & Data

LlamaParse

parse and extract only

Unstructured

ACL-aware connectors

Firecrawl

AGPL server · cloud with ZDR

Google Cloud

Google Document AI

processor region confirmed

Reducto

hard documents

Mistral OCR 4.1

pin the model ID

Apify

public data only

Crawl4AI

pre-1.0 · pilots

MinerU

licence thresholds apply

EVALUATION & OBSERVABILITY PLANE

L9 joins C8 as one evidence plane with two owners

L9

Evaluation & observability plane
evidence for every layer, from day one

One firm-owned OpenTelemetry spine, Git-versioned evaluation sets, one platform of record, and two red-team tools, one independent of the model vendor.

MLflow GenAI
where an ML platform exists

Arize AX
Dynatrace-owned

W&B Weave
CoreWeave-owned

Langfuse
ClickHouse-owned · self-host

DeepEval
Cl metric library

Braintrust
eval-led teams

LangSmith
LangGraph estates · BYOC, EU

Promptfoo
red-teaming · OpenAI deal announced

Datadog Agent Observability
Datadog APM estates

Opik
Comet · Apache-2.0

Arize Phoenix
Dynatrace-owned · ELv2

- **One gateway of record** for all model, tool and agent traffic: in region, deployed twice, failing closed.
- **Every agent a registered identity**, acting for a named person through short-lived tokens; deny by default.
- **One privacy service** at six points: ingestion, prompt, tool results, output, memory writes and trace export.
- **Git as the configuration of record**: prompts, model pins and tool lists released as one approved manifest.
- **Build order**: governance and evaluation, then models through the gateway, retrieval, workflows, tools — memory last.

- Tiers are for a regulated asset manager, scored on one eight-criterion rubric with regulated-FS weights. A Strategic tier carries its condition; the condition is the decision.
- Cloud-tagged items are alternatives chosen by primary cloud, not a shopping list.
- Disclosure: researched and drafted with an Anthropic model. Anthropic items were scored on the same rubric; their tiers were set by the reader, and an independent alternative is named for each.
- Personal research, not any firm's platform. Every tile is backed by sourced facts in the dataset and the product appendix.